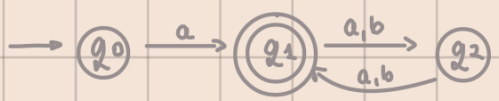
LFA

pregătire examen

Recapitulare 1:

☒ $L = \{ w \in \{a,b\}^* \mid |w| = \text{impară}, w \text{ începe cu } a \}$



$S \to aS_1$

$S_1 \to aS_2 \mid bS_2 \mid e$

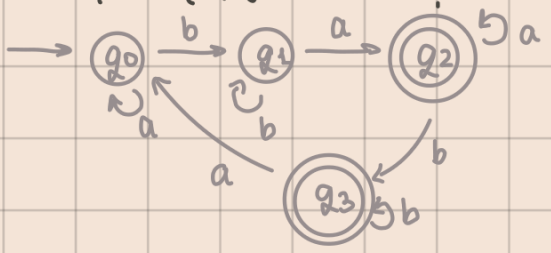
$S_2 \to aS_1 \mid bS_1$

$S = aS_1 = a((a+b)(a+b)^*$

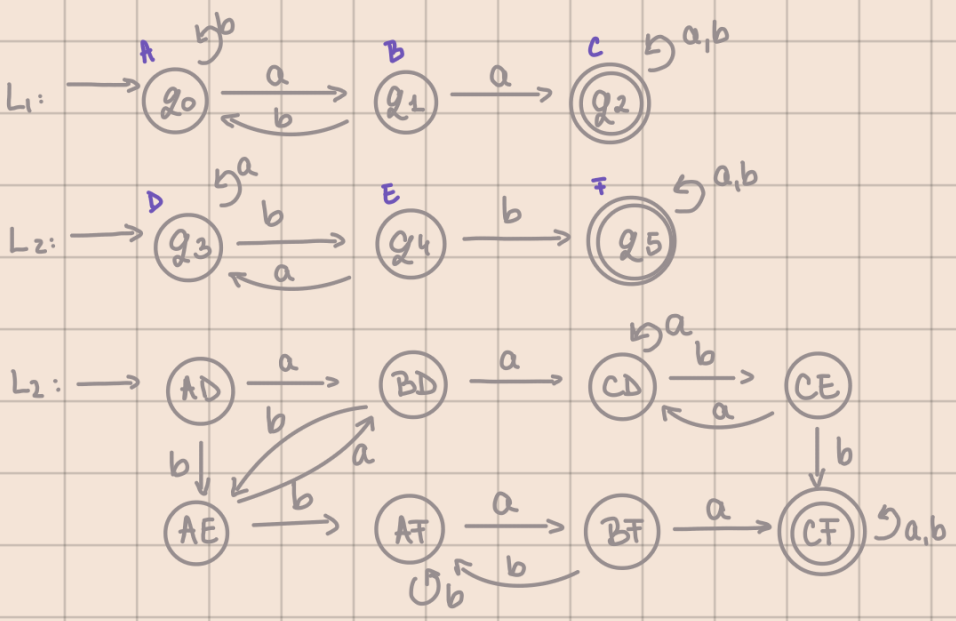
$S_1 = aS_2 + bS_2 + e = (a+b)(aS_1 + bS_1) + e = ((a+b)(a+b))^*$

$S_2 = aS_1 + bS_1$

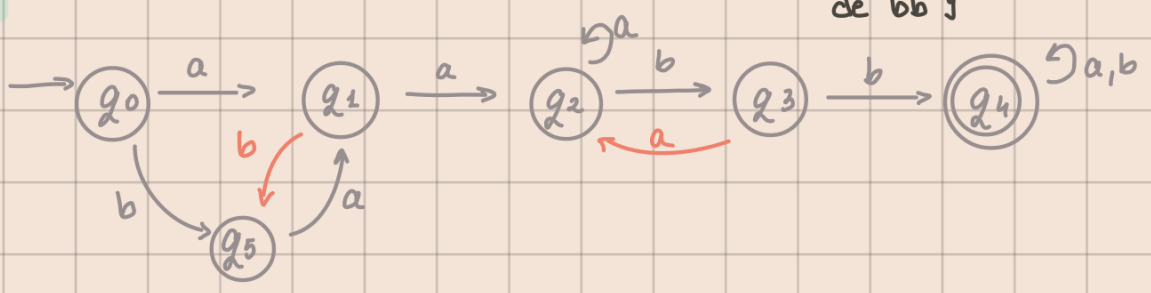
☒ $L = \{ w \in \{a,b\}^* \mid w \text{ conține un nr impar de ba} \}$ AFD



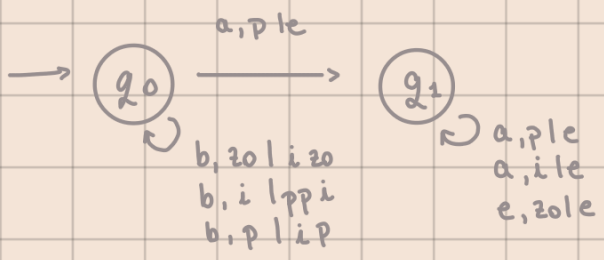
☒ $L = \{ w \in \{a,b\}^* \mid w \text{ conține aa și bb} \}$



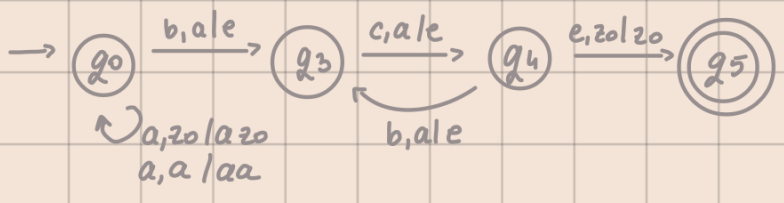
☒ $L = \{ w \in \{a,b\}^* \mid w$ conține aa și bb, prima apariție a lui aa înainte de bb $\}$



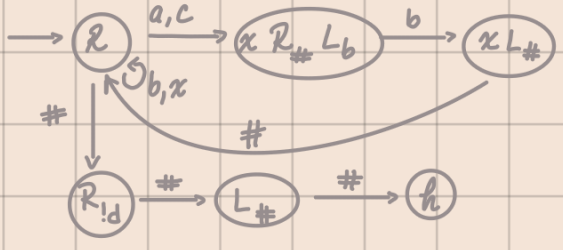
☒ $L = \{ b^{2n} a^{3n} \mid n \geq 1 \}$ LIC => AS



☒ $L = \{ a^{2n} (bc)^n \mid n \geq 1 \}$ LIC => AS



☒ $L = \{ w \in \{a,b,c\}^* \mid \#a(w) + \#c(w) = \#b(w) \}$ MT care acceptă



Recapitulare 2:

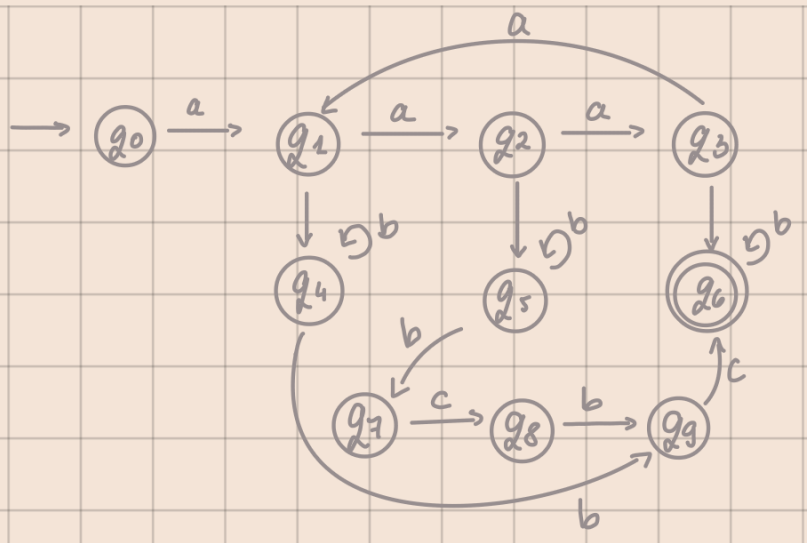
☒ $L = \{ a^n b^m (bc)^{n\%3} \mid m,n > 0 \}$

$S \to aS_1$

$S_1 \to aS_2 \mid bS_4$ $\quad S_4 \to bS_4 \mid bc$

$S_2 \to aS_3 \mid bS_5$ $\quad S_5 \to bS_5 \mid bcbc$

$S_3 \to aS_1 \mid bS_6$ $\quad S_6 \to bS_6 \mid e$



AFN

☒ $L = \{ w \in \{a,b\}^* \mid$ primul sb. ≠ ultimul sb. și #w = impar $\}$

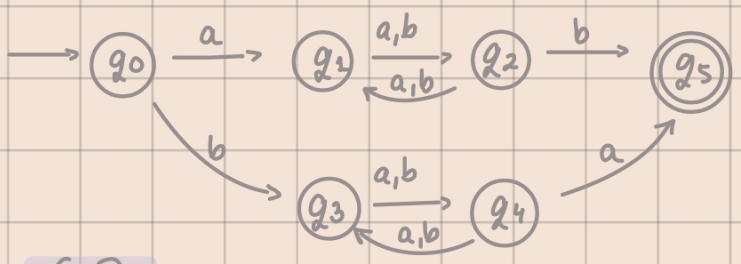
$S \to aS_1 \mid bS_2$

$S_1 \to aS_3 \mid bS_3$

$S_3 \to aS_1 \mid bS_1 \mid b$

$S_2 \to aS_4 \mid bS_4$

$S_4 \to aS_2 \mid bS_2 \mid a$ G.R.

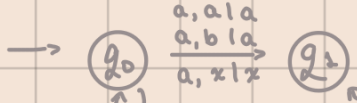


☒ $L = \{ w \in \{a,b\}^* \mid$ primul sb. ≠ ultimul sb. , #(w) = impară , sb. mij. este a $\}$

L.R. L.i.C

⇒ L.i.C ⇒ AS

sau: AS pornind de la GiC.



☒ $L = \{ ww \mid w \in \{a,b\}^*$, #(w) % 2 = 0 , w conține pe poz. pare a și pe poz. impare b $\}$

$S \to babaS \mid e$

☒ $L = \{ ww^R \mid w \in \{a\}^+$, #(w) % 2 = 1 $\}$

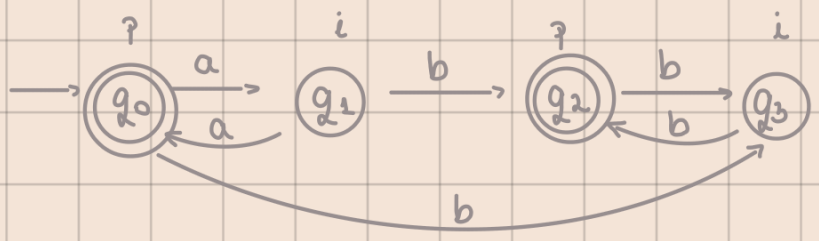
$S \to aaaaS \mid e$

$|wwR| = 2k+1 + 2k+1 = 4k+2 = M4+2$

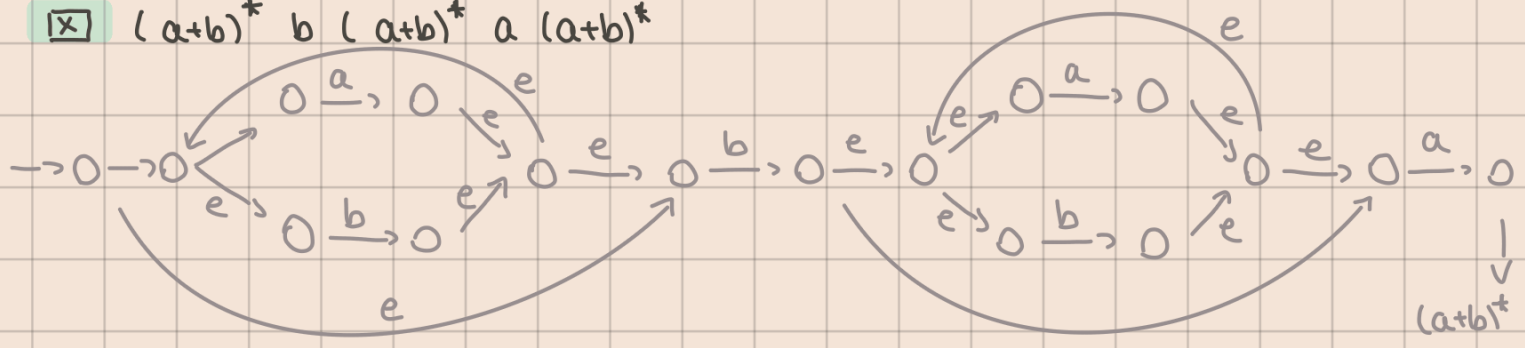
☒ $L = \{ a^n b^m \mid n - m = 2 \}$

$S \to aSb \mid aa$

☒ AFD minim pt. $\{ a^n b^m \mid n+m = par \}$



☒ $(a+b)^* \; b \; (a+b)^* \; a \; (a+b)^*$



☒ M.T. care calculează: $f: \{a,b\}^* \to \{a,b\}^*$ $f(w) = w'$

w' = toate aparițiile subșirului ab din w sunt înlocuite cu bb.

Recapitulare 3:

☒ Gram: $L = \{ a^i b^m c^n \mid i,m,n \geq 1 \;,\; n = |i-m| \}$

$S \to S_1 \mid S_2$ $\quad i \geq m \quad i = m+n$

$S \to aS_1c \mid aS_3c$ $\quad i < m \quad m = i+n$

$S_3 \to aS_3b \mid ab$

$S_2 \to S_4S_5$

$S_4 \to aS_4b \mid ab$

$S_5 \to bS_5c \mid bc$

☒ $L = \{ xy \mid x,y \in \{a,b\}^+ , \#(x) = \#y \}$

$S \to aaS \mid abS \mid baS \mid bbS \mid aa \mid ab \mid ba \mid bb$

☒ GR → AFD ⟸ VAR 1: GR → ER → AFD; VAR 2: înlocuiri

$S \to aB \mid bS$ $\quad S \to aB \mid bS$

$B \to S \mid a \mid b$ $\quad B \to aB \mid bS \mid a \mid b$



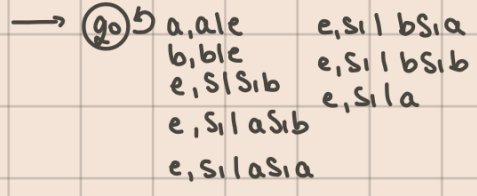
PREGĂTIRE 2023 - 2024

1) Scrieți o gram. pt $L = \{ uavb \mid u, v \in \{a,b\}^*, \#(u) = \#(v) \}$

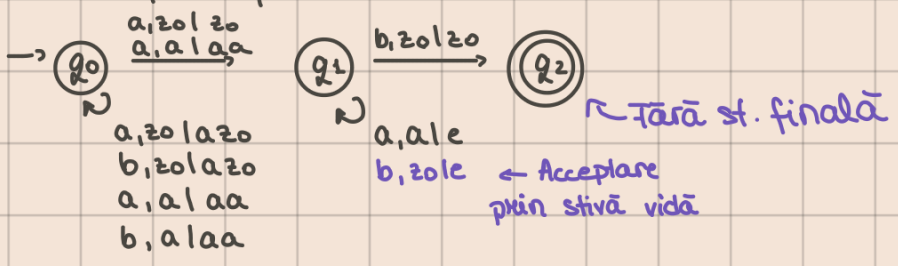
$S \to S_1 b$

$S_1 \to aS_1a \mid aS_1b \mid bS_1a \mid bS_1b \mid a$ GIC

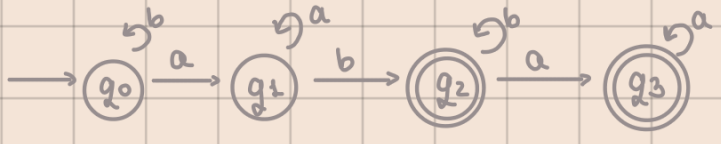
Automat cu stivă



↳ Acceptare prin stivă vidă



2) $L = \{ w \in \{a,b\}^* \mid w$ conține ab o singură dată $\}$



Gramatică: $S \to bS \mid aS_1$

$S_1 \to aS_1 \mid bS_2$

$S_2 \to bS_2 \mid aS_3 \mid e$

$S_3 \to aS_3 \mid e$

3) GR care gen. lb. descrisă de ER $(a+b)^* a^* b^+$

$\Delta = (a+b)^* a^* b^+ = (a+b)\Delta + \underbrace{a^* b^+}_{\Delta_1}$

$\Delta_1 = a\Delta_1 + \underbrace{b^+}_{\Delta_2}$

$\Delta_2 = b^* b = b\Delta_2 + b$

$S \to aS \mid bS \mid S_1$

$S_1 \to aS_1 \mid S_2$

$S_2 \to bS_2 \mid b$

4) AFD pt lb. descrisă prin ER: $(aa+b)^*(a+b)^+b^*\#$
(1 2 3 4 5 6 7)

sb	index	follow-pos
a	1	2
a	2	3,1,4,5
b	3	1,3,4,5
a	4	4,5,6,7
b	5	4,5,6,7
b	6	6,7
#	7	—

Q_0 = firstpos(ER) = $\{1,3,4,5\}$

$m(Q_0,a) = fp(\{1,4\}) = \{2,4,5,6,7\} = Q_1$

$m(Q_0,b) = fp(\{3,5\}) = \{1,3,4,5,6,7\} = Q_2$

$m(Q_1,a) = fp(\{2,4\}) = \{1,3,4,5,6,7\} = Q_2$

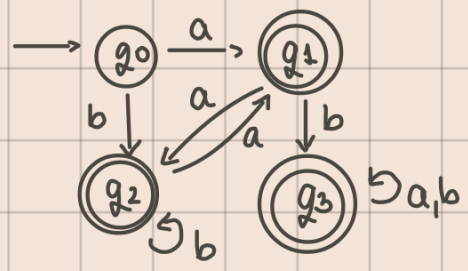
$m(Q_1,b) = fp(\{5,6\}) = \{4,5,6,7\} = Q_3$

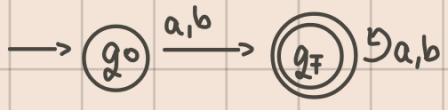
$m(Q_2,a) = fp(\{1,4\}) = Q_1$

$m(Q_2,b) = fp(\{3,5,6\}) = \{1,3,4,5,6,7\} = Q_2$

$m(Q_3,a) = fp(\{4\}) = \{4,5,6,7\} = Q_3$

$m(Q_3,b) = fp(\{5,6\}) = Q_3$





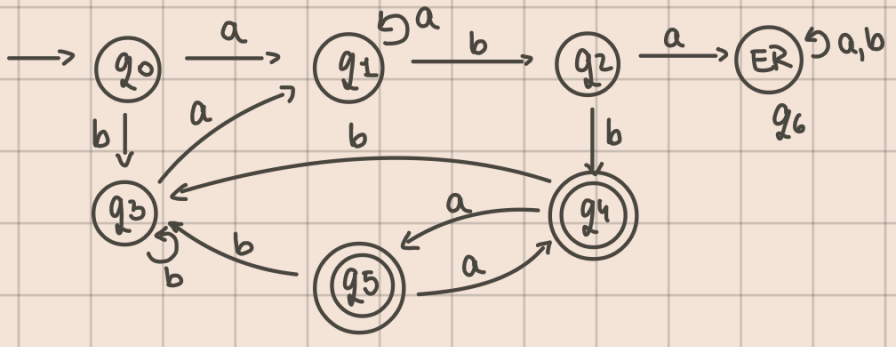
$P = \{q_0\}$ $F = \{q_1, q_2, q_3\}$

$m(q_1,a) = q_2 \in F$ $m(q_1,b) = q_3 \in F$

$m(q_2,a) = q_2 \in F$ $m(q_2,b) = q_2 \in F$ $\Rightarrow q_1 \equiv q_2 \equiv q_3$

$m(q_3,a) = q_3 \in F$ $m(q_3,b) = q_3 \in F$

5) Minimizați AFD:



$P = \{q_0, q_1, q_2, q_3, q_6\}$ $F = \{q_4, q_5\}$

stări echivalente pt a

$m(q_0,a) = q_1 \in P_{22}$

$m(q_1,a) = q_1 \in P_{22}$

$m(q_2,a) = q_6 \in P$

$m(q_3,a) = q_1 \in P_{22}$

$m(q_6,a) = q_6 \in P_{21}$

$m(q_0, b) = q_3 \in P_{211}$

$m(q_1, b) = q_2 \in P_1$

$m(q_2, b) = q_4 \in F$

$m(q_3, b) = q_3 \in P_{211}$

$m(q_6, b) = q_6 \in P_2$

$P = P_1 \cup P_2$

$P_2 = \{q_0, q_1, q_3, q_6\}$

$P_1 = \{q_2\}$

$P_2 = P_{21} \cup P_{22}$

$P_{21} = \{q_0, q_3, q_6\}$

$P_{22} = \{q_1\}$

$P_{21} = P_{211} \cup P_{212}$ $\quad P_{211} = \{q_0, q_3\}$ $\quad P_{212} = \{q_6\}$

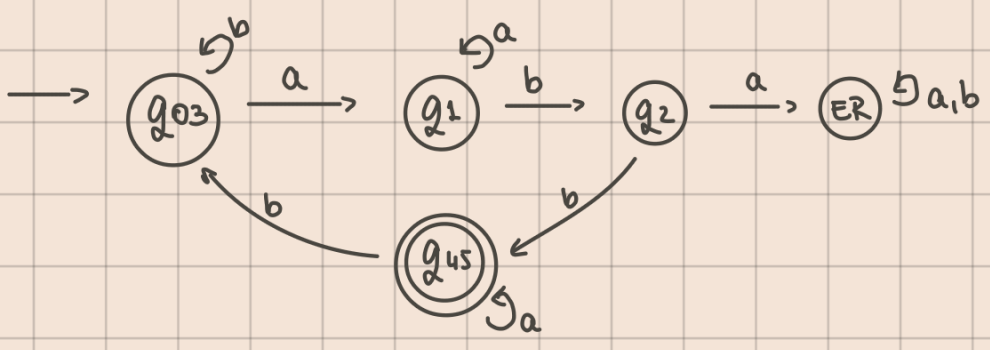
$m(q_4, a) = q_5 \in F$

$m(q_5, a) = q_5 \in F$

$m(q_4, b) = q_3 \in P_{211}$

$m(q_5, b) = q_3 \in P_{211}$

$q_4 \equiv q_5$



X . . .

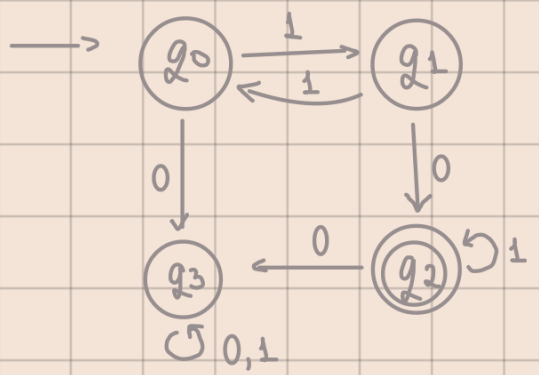
EXAMEN 2023:

① Gram: $L = \{ uww^Rv \mid u,v,w \in \{a,b\}^+ \}$

$S \to aS \mid bS \mid aaaS_1 \mid baaS_1 \mid abbS_1 \mid bbbS_1$

$S_1 \to aS_1 \mid bS_1 \mid a \mid b$

② ER pt: $m(q_0,1) = q_1$, $m(q_0,0) = q_3$, $m(q_1,0) = q_2$, $m(q_1,1) = q_0$, $m(q_2,1) = q_2$, $m(q_2,0) = q_3$, $m(q_3,0) = q_3$, $m(q_3,1) = q_3$



$S \to 1S_1 \mid 0S_3$

$S_1 \to 1S \mid 0S_2$

$S_2 \to 1S_2 \mid 0S_3 \mid e$

$S_3 \to 0S_3 \mid 1S_3$

$\Delta = 1\Delta_1 + 0\Delta_3 = 11\Delta + 101^*(0(0+1)^* + e) + 0(0+1)^*$

$\Delta_1 = 1\Delta + 0\Delta_2 = 1\Delta + 01^*(0(0+1)^* + e)$

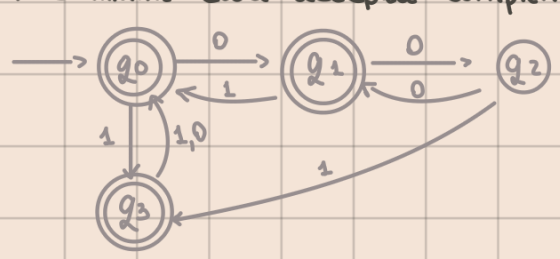
$\Delta_2 = 1\Delta_2 + 0\Delta_3 + e = 1^*(0(0+1)^* + e)$

$\Delta_3 = 0\Delta_3 + 1\Delta_3 = (0+1)^*$

$\Delta = (11)^*(101^*(0(0+1)^* + e) + 0(0+1)^*)$

③ AFD minim care acceptă complem. $L = \{ w \in \{0,1\}^* \mid w$ se term. cu 00 și #(w)-pară $\}$

$\bar{L}$:



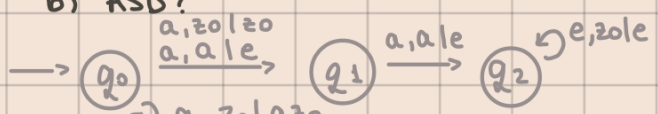
④ $L = \{ w \in \{a,b\}^* \mid \#a(w) = \#b(w) + 2 \}$

a) Gram $\Rightarrow$ AS

$S \to S_1 aS_1 aS_1$

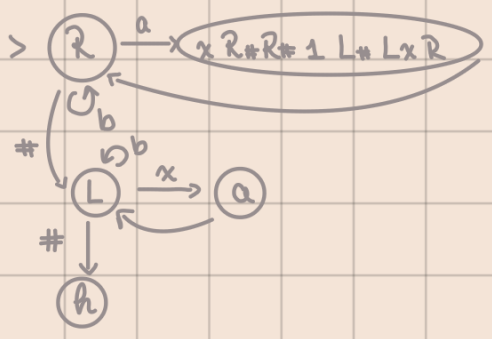
$S_1 \to aS_1bS_1 \mid bS_1aS_1 \mid e$

b) ASD?



X

⑤ $\#w\# \Rightarrow \#w\#n\#$, $n = \#a(w)$ n sb. de 1 pe bandă



VAR. 2 EXAMEN:

① $L = \{ w \in \{a,b\}^* \mid w = uxu^R , u,x \in \{a,b\}^+ , \#(w) = \text{imp} \}$

$S \to aS_1 \mid bS_2$

$S_1 \to aS_3 \mid bS_3 \mid aa \mid ba \qquad S_3 \to aS_1 \mid bS_1$

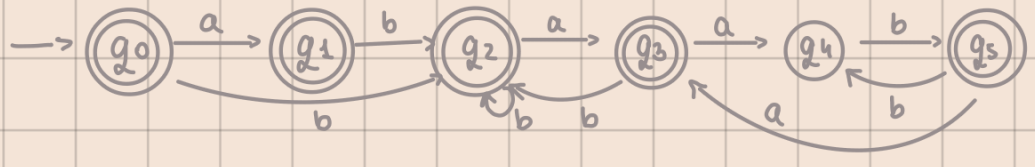
$S_2 \to aS_4 \mid bS_4 \mid ab \mid bb \qquad S_4 \to aS_2 \mid bS_2$ GR

② $L = \{ w \in \{a,b\}^* \mid \#a(w) = \#b(w) \}$

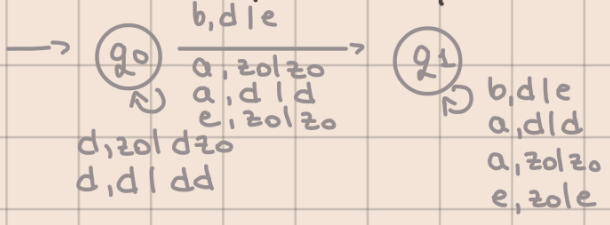
↳ LIC

$L' \subset L = LR : L' = \{ (ab)^n \mid n \geq 0 \}$

③ ~~AFD~~ minim care acceptă $L = \{ w \in \{a,b\}^* \mid$ fiecare grup din exact 2 sb de a consec. e format de un nr. impar de b-uri , w nu începe cu un astfel de grup $\}$



④ AS care acceptă $L = \{ d^p w \mid w \in \{a,b\}^* , p = \#b(w) \}$



? ⊠ AS care acceptă $L = \{ w_1 c w_2 c \dots c w_k c x \mid x, w_{1\dots k} \in \{a,b\}^* , k \in \mathbb{N} ,$ $x = w_j^R$

VAR. 3 EXAMEN:

① $L = \{ w \in \{a,b\}^* \mid w = a^n a^m b^p b^n,\ \#(w) = \text{imp.} \}$

$S \to aaS \mid aS_1 \mid bS_1$

$S_1 \to bbS_1 \mid e$

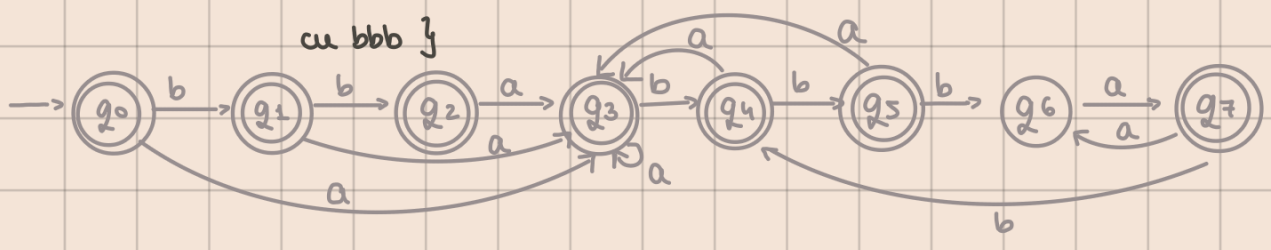
② $L = \{ a^n b^m \mid n < m \}$

$S \to aSb \mid bS_1$

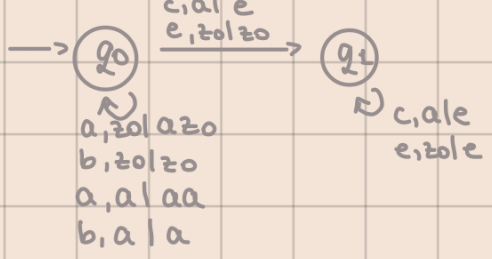
$S_1 \to bS_1 \mid e$ LiC

$L' \subset L = GR \quad L' = \{ b^m \mid m > 0 \}$

③ AFD minim: $L = \{ w \in \{a,b\}^* \mid bbb$ este urm. de nr. imp. de a-uri, nu începe cu $bbb \}$

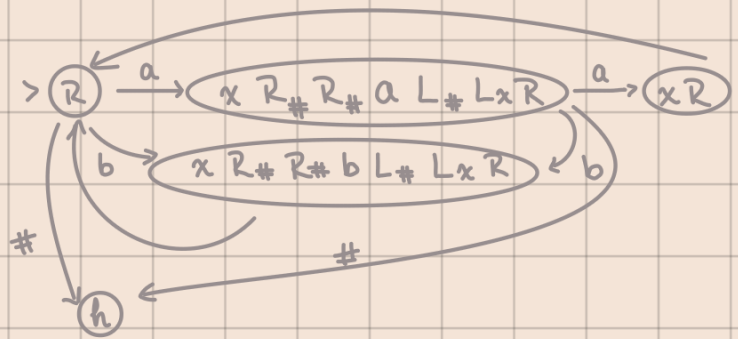


④ AS: $L = \{ wc^k \mid w \in \{a,b\}^*,\ k = \#a(w) \}$

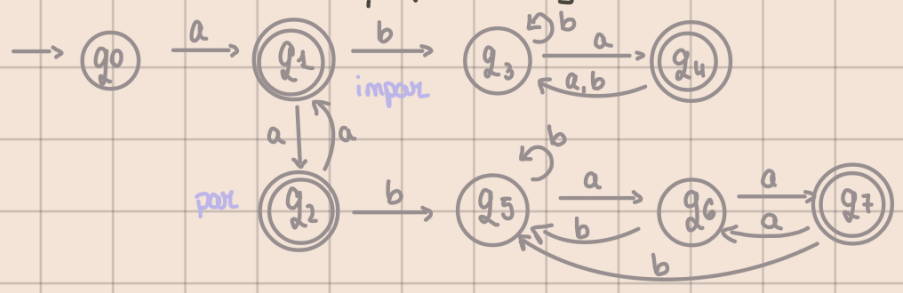


⑤ MTC: $f: \{a,b\}^* \to \{a,b\}^* \quad f(w) = w'$

w' = orice aa devine a



☒ AFD minim : $L = \{ w \in \{a,b\}^* \mid w$ începe și se term. cu a, cel mai lung sufix de a și cel mai lung prefix de a au lungimea de aceeași paritate $\}$



☒ Gram. pt : $L = \{ ww \mid w \in \{a,b\}^+$ și w conține pe poz. pare a și pe imp. b$\}$

baba babbab

$S \rightarrow S_1 \mid S_2 b$

$S_1 \rightarrow babaS_1 \mid \epsilon$

$S_2 \rightarrow baSba \mid b$